QUALIFICATION
REQUIREMENTS
REMOVED

MIL-M-38510/490A
11 September 1986
SUPERSEDING
MIL-M-38510/490(USAF)
1December 1981

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE,
MONOLITHIC, 8-BIT MICROCOMPUTER

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic N-channel, silicon gate microcomputer. Two product assurance classes and a choice of case outline/lead finish are provided for each type, and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	8-bit microcomputer, masked programmable ROM (1K x 8)
02	8-bit microcomputer (without ROM)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designed as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
Q	D-5 (40-lead, 9/16" x 2-1/16") dual-in-line package

1.3 Absolute maximum ratings.

V_{DD} and V_{CC} supply voltage range	-0.5 to +7.0 Vdc
Voltage on any pin (referenced to ground)	-0.5 to +7.0 Vdc
Storage temperature range	-65° to +150°C
Maximum power dissipation of package for outline shown above	1.5 W
Maximum device power dissipation	
T_C = -55°C	0.7 W
T_C = +125°C	0.5 W
Lead temperature (soldering 5 seconds)	270°C
Maximum junction temperature (T_J) 1/	175°C
Thermal resistance, junction to case (θ_{JC})	(See MIL-M-38510, appendix C)

1/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC} and V_{DD})	4.5 to 5.5 Vdc
Minimum high-level input voltage 2/	
(Reset, XTAL1, XTAL2)	3.8 V
(All other inputs)	2.3 V
Maximum low-level input voltage 2/	
(Reset, XTAL1, XTAL2)	0.5 V
(All other inputs)	0.7 V
Crystal input frequency	1.0 to 6.0 MHz
Case operating temperature range	-55° to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILTARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510 and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.4.

2/ The following conditions must be satisfied when using the T1 input:
The maximum rate at which the event counter may be incremented is once per three instruction cycles (every 7.5 µs when the crystal frequency is 6 MHz).
The T1 input must remain high for at least 500 ns (at 6 MHz) after each transition.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55^{\circ}C \leq T_C \leq +125^{\circ}C$	Figure	Limits Min	Limits Max	Unit
High level output voltage (BUS)	V_{OH1}	$V_{CC} = V_{DD} = 4.5$ V $I_{OH} = -240$ µA		2.4		V
High level output voltage ($\overline{RD}$,$\overline{WR}$,$\overline{PSEN}$,ALE)	V_{OH2}	$V_{CC} = V_{DD} = 4.5$ V $I_{OH} = -50$ µA		2.4		V
High level output voltage (P10-17,P20-27,$\overline{PROG}$,T0)	V_{OH3}	$V_{CC} = V_{DD} = 4.5$ V $I_{OH} = -30$ µA		2.4		V
Low level output voltage (BUS,$\overline{RD}$,$\overline{WR}$,$\overline{PSEN}$,ALE)	V_{OL1}	$V_{CC} = V_{DD} = 4.5$ V $I_{OL} = 1.2$ mA			.45	V
Low level output voltage (P10-17,P20-27,$\overline{PROG}$,T0)	V_{OL2}	$V_{CC} = V_{DD} = 4.5$ V $I_{OL} = 0.8$ mA			.45	V
High level input current ($\overline{RESET}$)	I_{IH4}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = 5.5$ V			-700	µA
Low level input current ($\overline{RESET}$)	I_{IL4}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = .45$ V			-700	µA
High level input current (T1,$\overline{INT}$, BUS,T0)	I_{IH1}	$V_{DD} = 5.5$ V $V_{CC} = 5.5$ V $V_{in} = 5.5$ V		-10	10	µA
High level input current (P10-17,P20-27,EA,$\overline{SS}$)	I_{IH2}	$V_{DD} = 5.5$ V $V_{CC} = 5.5$ V $V_{in} = 5.5$ V			-700	µA
Low level input current (P10-17,P20-27,EA,SS)	I_{IL2}	$V_{DD} = 5.5$ V $V_{CC} = 5.5$ V $V_{in} = .45$ V			-700	µA
Low level input current (T1,$\overline{INT}$, BUS,T0)	I_{IL1}	$V_{DD} = 5.5$ V $V_{CC} = 5.5$ V $V_{in} = 5.5$ V		-10	10	µA
High level input current (XTAL1)	I_{IH3}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = 5.5$ V		-50	50	µA
Low level input current (XTAL1)	I_{IL3}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = 0.45$ V		-50	50	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Figure	Limits Min	Limits Max	Unit
Input capacitance (Port 1, Port 2, BUS, T0, T1, $\overline{INT}$, $\overline{RESET}$, $\overline{SS}$, EA, XTAL1, XTAL2)	C_I	$V_{CC} = V_{DD} = V_{SS} = 0$ V f = 1 MHz		2	30	pF
Output capacitance ($\overline{PROG}$, $\overline{RD}$, $\overline{WR}$, ALE, $\overline{PSEN}$)	C_O	$V_{CC} = V_{DD} = V_{SS} = 0$ V f = 1 MHZ		2	20	pF
Input clamp voltage	V_{IC}	$I_O = -200$ µA			-4.0	V
V_{DD} supply current	I_{DD1}	$V_{CC} = 5.5$ V, $V_{DD} = 5.5$ V			10	mA
	I_{DD2}	$V_{CC} = 0$ V, $V_{SS} = 0$ V			10	mA
V_{CC} supply current	I_{CC}	$V_{CC} = V_{DD} = 5.5$ V $V_{SS} = 0$ V			90	mA
High level input current (XTAL2)	I_{IH5}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = 5.5$ V			13	mA
Low level input current (XTAL2)	I_{IL5}	$V_{CC} = V_{DD} = 5.5$ V $V_{in} = 0.45$ V			13	mA
Clock cycle time (T0) 2/	t_{CLK}	1/		0.5	3	µs
Clock pulse width (T0)	t_{PWH8}	1/		56		ns
ALE pulse width	t_{PWH1}	1/	6	280		ns
Address valid to ALE (↓)	t_{PHL1}	1/	6	290		ns
ALE (↓) to BUS floating	t_{PHL2} t_{PLH2}	1/	6	20		ns
$\overline{PSEN}$ pulse width	t_{PWL2}	1/	6	600		ns
$\overline{RD}$ pulse width	t_{PWL3}	1/	7	950		ns
$\overline{WR}$ pulse width	t_{PWL4}	1/	8	950		ns
Data valid to $\overline{WR}$ (↑)	t_{PLH3}	1/	8	730		ns
$\overline{WR}$ (↑) to BUS floating	t_{PHL4} t_{PLH4}	1/	8	10		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Figure	Limits Min	Limits Max	Unit
ALE cycle time 3/	t_{cy}	1/	6	2.5	15	µs
$\overline{RD}$ (↑) to data hold time	t_{HHL1} t_{HLH1}	1/	7	0	270	ns
Data valid to $\overline{RD}$ setup time	t_{SLH1}	1/ 4/	7		280	ns
BUS floating to $\overline{RD}$ (↑)	t_{PHL5}	1/	7	190		ns
Address valid to $\overline{WR}$ (↓)	t_{PHL7}	1/	8	500		ns
Address valid to data valid setup time	t_{SHL3} t_{SLH3}	1/	7		1330	ns
Port control to $\overline{PROG}$ (↓)	t_{PHL9}	1/	9	120		ns
$\overline{PROG}$ (↓) to port control invalid	t_{PHL10} t_{PLH10}	1/	9	360		ns
Input data valid to $\overline{PROG}$ setup time	t_{SLH5}	1/ 4/	9		1020	ns
PROG (↑) to input data valid hold time	t_{HHL5} t_{HLH5}	1/	9	0	270	ns
Output data to $\overline{PROG}$ (↑)	t_{PLH11}	1/	9	700		ns
$\overline{PROG}$ (↑) to output data invalid	t_{PHL12} t_{PLH12}	1/	9	100		ns
$\overline{PROG}$ pulse width	t_{PWL5}	1/	9	1250		ns
P20-23 input data to ALE (↑) setup time	t_{SLH6}	1/	10	360		ns
ALE (↑) to P20-23 input data hold time	t_{HHL6} t_{HLH6}	1/	10	50		ns
$\overline{PSEN}$, $\overline{WR}$, $\overline{RD}$, and $\overline{PROG}$, to ALE (↑)	t_{PLH8}	1/	6, 7, 8, 9	10		ns
$\overline{RESET}$ pulse width (initialization)	t_{PWL6}	$V_{CC} = V_{DD} = 4.5$ V min.	11	10		ms

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55^{\circ}C \leq T_C \leq +125^{\circ}C$	Figure	Limits Min	Limits Max	Unit
$\overline{\text{RESET}}$ pulse width (stabilized oscillator)	t_{PWL7}	XTAL frequency = 6 MHz 5/	12	12.5		µs
PSEN (↑) to instruction hold time	t_{HHL2} t_{HLH2}	1/	6	0	270	ns
Instruction valid to $\overline{\text{PSEN}}$ setup time	t_{SLH2}	1/ 4/	6	400		ns
BUS floating to $\overline{\text{PSEN}}$ (↓)	t_{PHL6}	1/	6	-40		ns
Address valid to instruction valid setup time	t_{SHL4} t_{SLH4}	1/	6		830	ns

1/ The load shall be as specified on figure 9.

2/ Clock frequency TO is 1/3 of crystal frequency.

3/ ALE frequency is 1/15 of crystal frequency.

4/ The true definitions of three of the setup times are given below. The definitions given in table I and figures 10, 11, and 13 are used to facilitate testing of these parameters.

a. Data valid to $\overline{\text{RD}}$ setup time = $t_{PWL3} - t_{SLH1}$

b. Instruction valid to $\overline{\text{PSEN}}$ setup itme = $t_{PWL2} - t_{SLH2}$

c. Input data valid to $\overline{\text{PROG}}$ setup time = $t_{PWL5} - t_{PSLH5}$

5/ The required $\overline{\text{RESET}}$ pulse width if power is on and the oscillator has stabilized is 5 machine cycles (12.5 µs at 6 MHz).

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3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II for the applicable device type and device class. The subgroups of table IIIC which constitute the minimum electrical test requirements for screening, qualification, and qualtiy conformance by device class are specified in table II.

3.6 Microcomputer instructions. Microcomputer instructions shall be as specified in table IV.

3.7 Microcircuit group assignment. The device covered by this specification shall be in microcircuit group number 107 (see MIL-M-38510, appendix E).

3.8 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The "JAN" or "J" certification mark shall not be used.

3.9 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.10 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit as shown on figure 4, or equivalent.

(2) The ambient temperature (T_A) shall be 125°C minimum.

b. Constant acceleration in accordance with test condition B of method 2001 of MIL-STD-883.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be as those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.3).

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroup 4 (C_I, C_O measurement) shall be performed only for initial qualification and after process or design changes which may affect input capacitance.

c. Subgroup 12 shall be performed only during initial qualification and after process or design changes. Variables data shall be taken.

d. The LTPD's per subgroup 10 of MIL-STD-883, method 5005, table I shall apply to table III, subgroup 12. Variables data shall be recorded for initial qualification at 125°C as a minimum.

e. Subgroup 7 and 8 functional testing shall include verification of the instruction set, table IV.

f. Subgroups 5 and 6 of table II of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. A special subgroup shall be added to the group B inspection requirements for class S and B devices and shall consist of the tests, conditions, and limits specified in 4.5.3. The LTPD for this subgroup shall be 15 for all classes.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III unless otherwise specified)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)(optional)	1,7	1,7
Final electrical test parameters (method 5004)	1,2,3,7,8, 1/	1,2,3,7,8 1/
Group A test requirements (method 5005)	1,2,3,7,8	1,2,3,4,7,8 9,10,11,12 (125°C)
Group B V_{ZAP} test	See 4.5.2	See 4.5.2
Group C end-point electrical parameters (method 5005)	N/A	1,2,3,8 (125°C)
Additional electrical subgroups for group C periodic inspections	N/A	12 2/
Group D end-point electrical parameters (method 5005)	1,2,3,7,8	1,2,3,8 (125°C)

1/ PDA applies to subgroups 1 and 7, (see 4.2c).

2/ Subgroup 12 is defined in table III. This subgroup shall be performed for initial qualification only and variables data shall be submitted. Subgroup 12 is sampled to an LTPD of 10 for class S and B, and to an LTPD of 15 for class C.

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Device types 01 and 02, top view

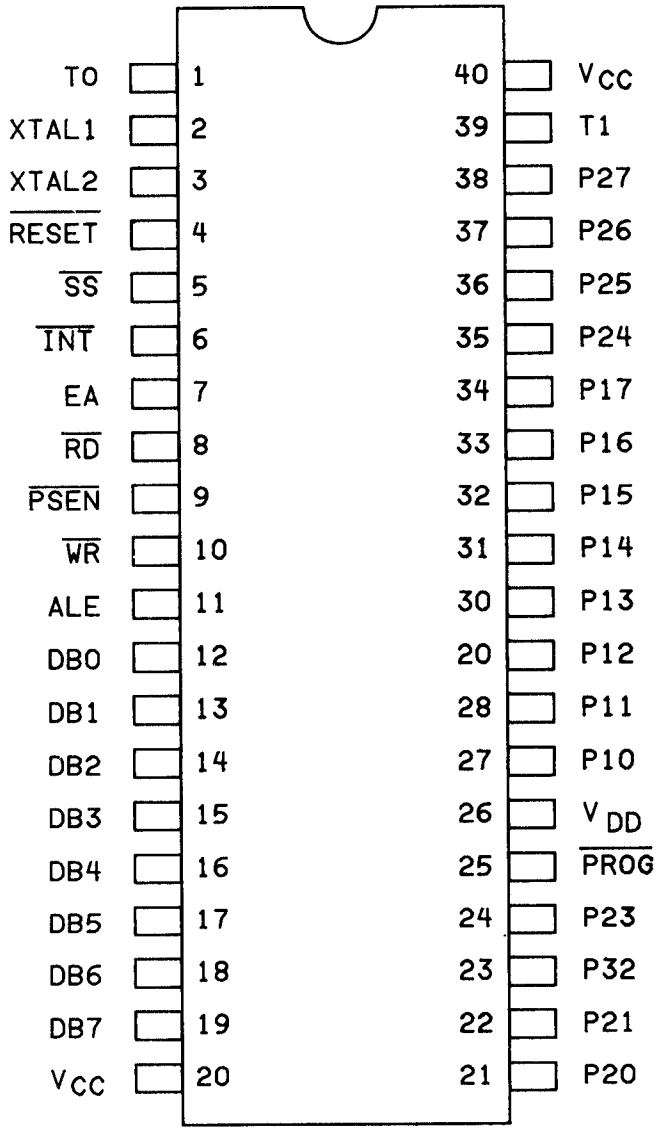


FIGURE 1. Terminal connections.

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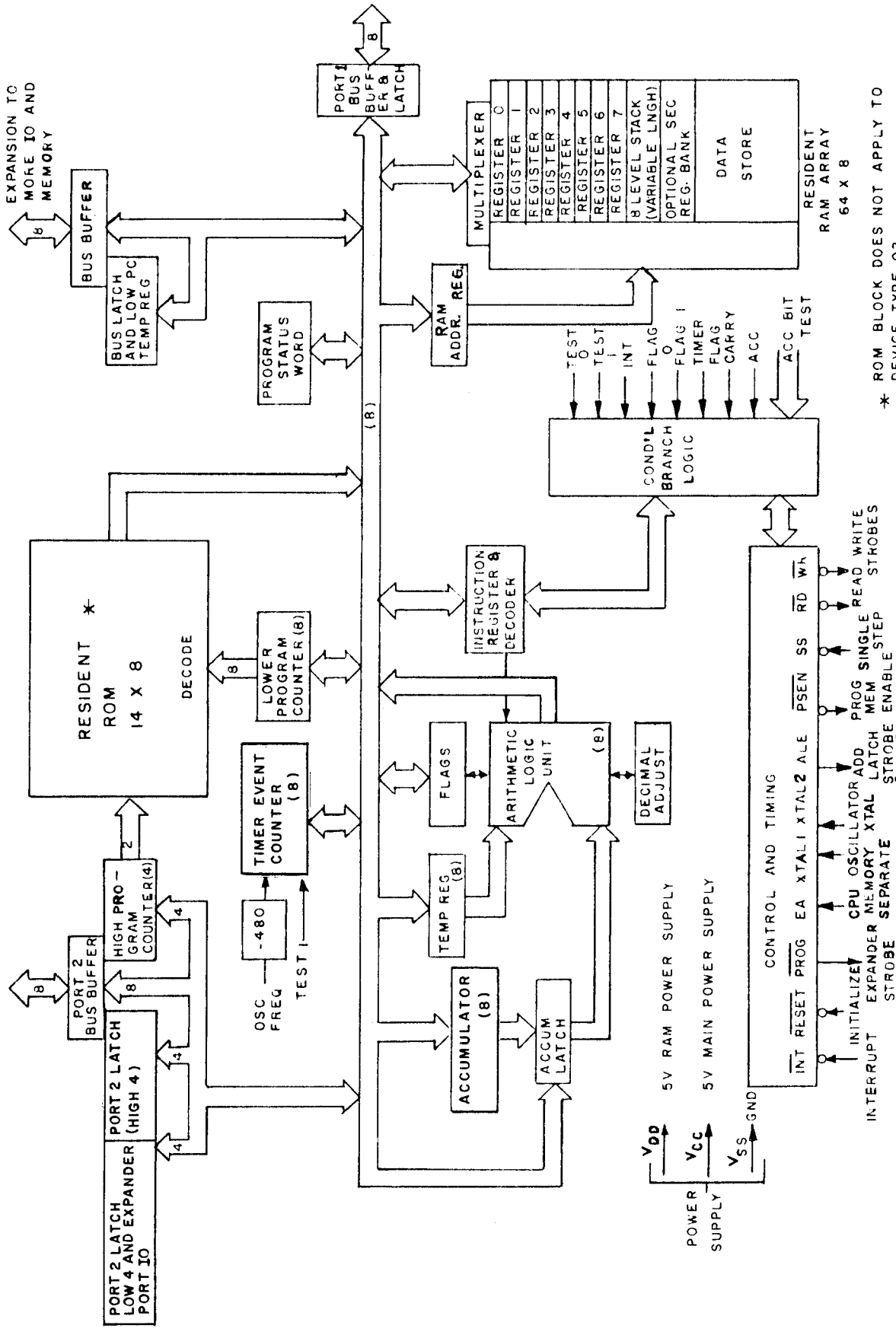
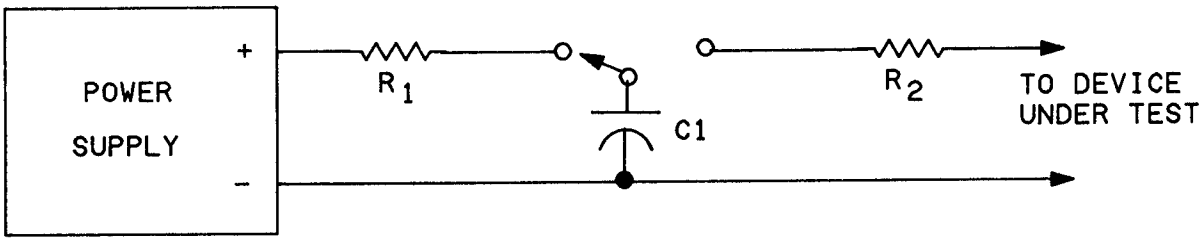


FIGURE 2. Functional block diagram.

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R_1 =appropriate current limiting resistor.

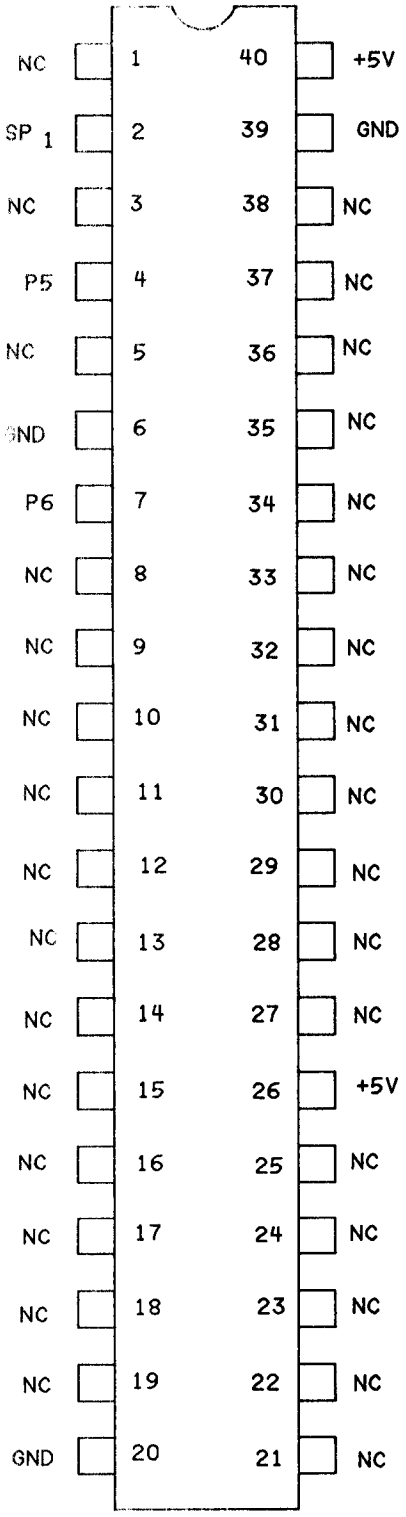
R_2 =1.5 kΩ ±5%

C1 =100 pF ± 20%

V_{ZAP} = 400 V. measured at device input terminal.

FIGURE 3. High voltage (V_{ZAP}) test circuit.

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PIN	PIN NAME	STRESS
1	TO	NC
2	X1	SP 1
3	X2	NC
4	$\overline{\text{RESET}}$	P 5
5	$\overline{\text{SS}}$	NC
6	$\overline{\text{INT}}$	GND
7	EA	P 6
8	$\overline{\text{RD}}$	NC
9	$\overline{\text{PSEN}}$	NC
10	$\overline{\text{WR}}$	NC
11	ALE	NC
12	DB0	NC
13	DB 1	NC
14	DB2	NC
15	DB3	NC
16	DB4	NC
17	DB5	NC
18	DB6	NC
19	DB7	NC
20	V_{SS}	GND
21	P20	NC
22	P21	NC
23	P22	NC
24	P23	NC
25	$\overline{\text{PROG}}$	NC
26	V_{DD}	+5V
27	P10	NC
28	P11	NC
29	P12	NC
30	P13	NC
31	P14	NC
32	P15	NC
33	P16	NC
34	P17	NC
35	P24	NC
36	P25	NC
37	P26	NC
38	P27	NC
39	T1	GND
40	V_{CC}	+5V

NOTE: All NC (not connected) pins shall remain open.

FIGURE 4. Burn-in and steady state life test circuit and waveforms.

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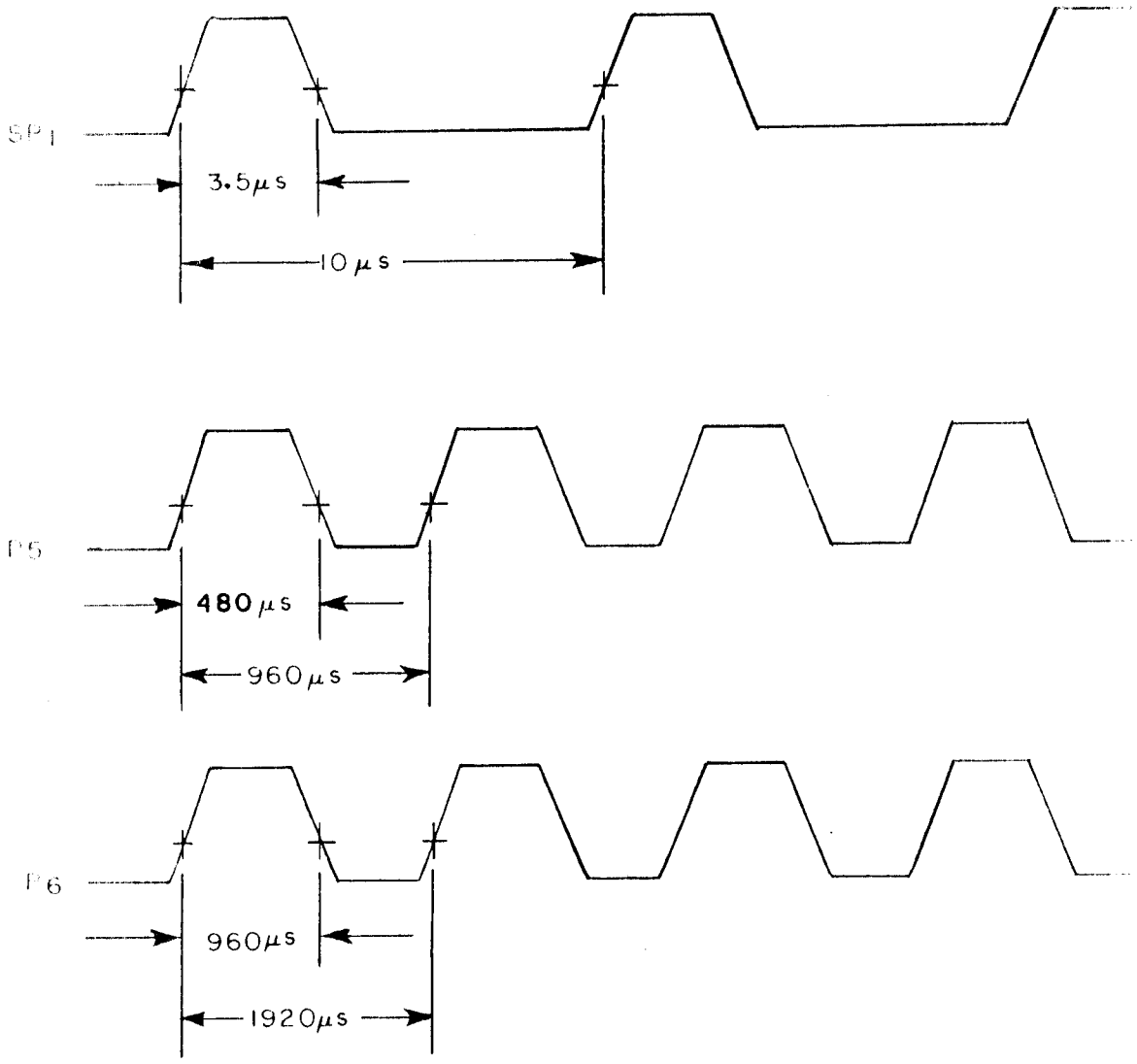
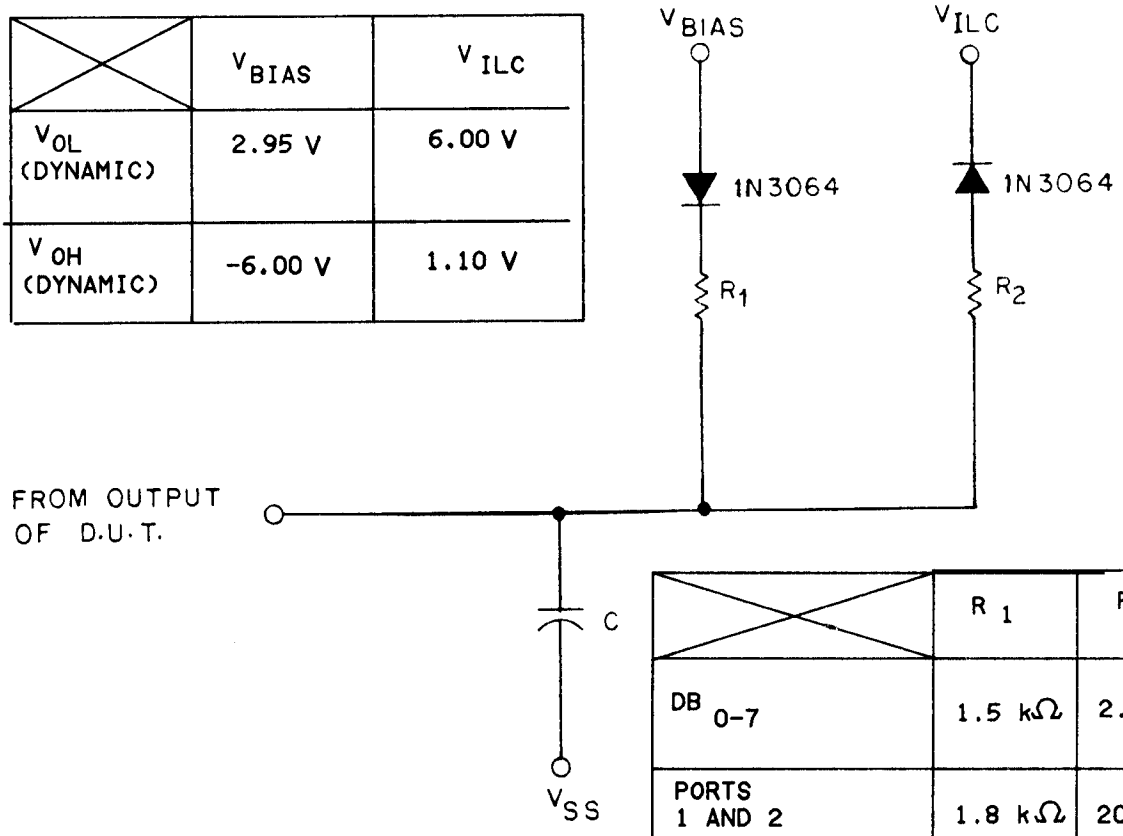


FIGURE 4. Burn-in and steady state life test circuit and waveforms - Continued.

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	V_{BIAS}	V_{ILC}
V_{OL} (DYNAMIC)	2.95 V	6.00 V
V_{OH} (DYNAMIC)	-6.00 V	1.10 V



	R_1	R_2
DB_{0-7}	1.5 kΩ	2.4 kΩ
PORTS 1 AND 2	1.8 kΩ	20 kΩ
PROG, $\overline{PSEN}$ $\overline{RD}$, $\overline{WR}$, ALE	1.5 kΩ	10 kΩ
TO	1.5 kΩ	10 kΩ

Capacitance (C) includes tester, wiring, and stray capacitance: 120 pF min for Bus; 80 pF min for control, ports 1 and 2.

FIGURE 5. Output load circuit for functional and switching tests.

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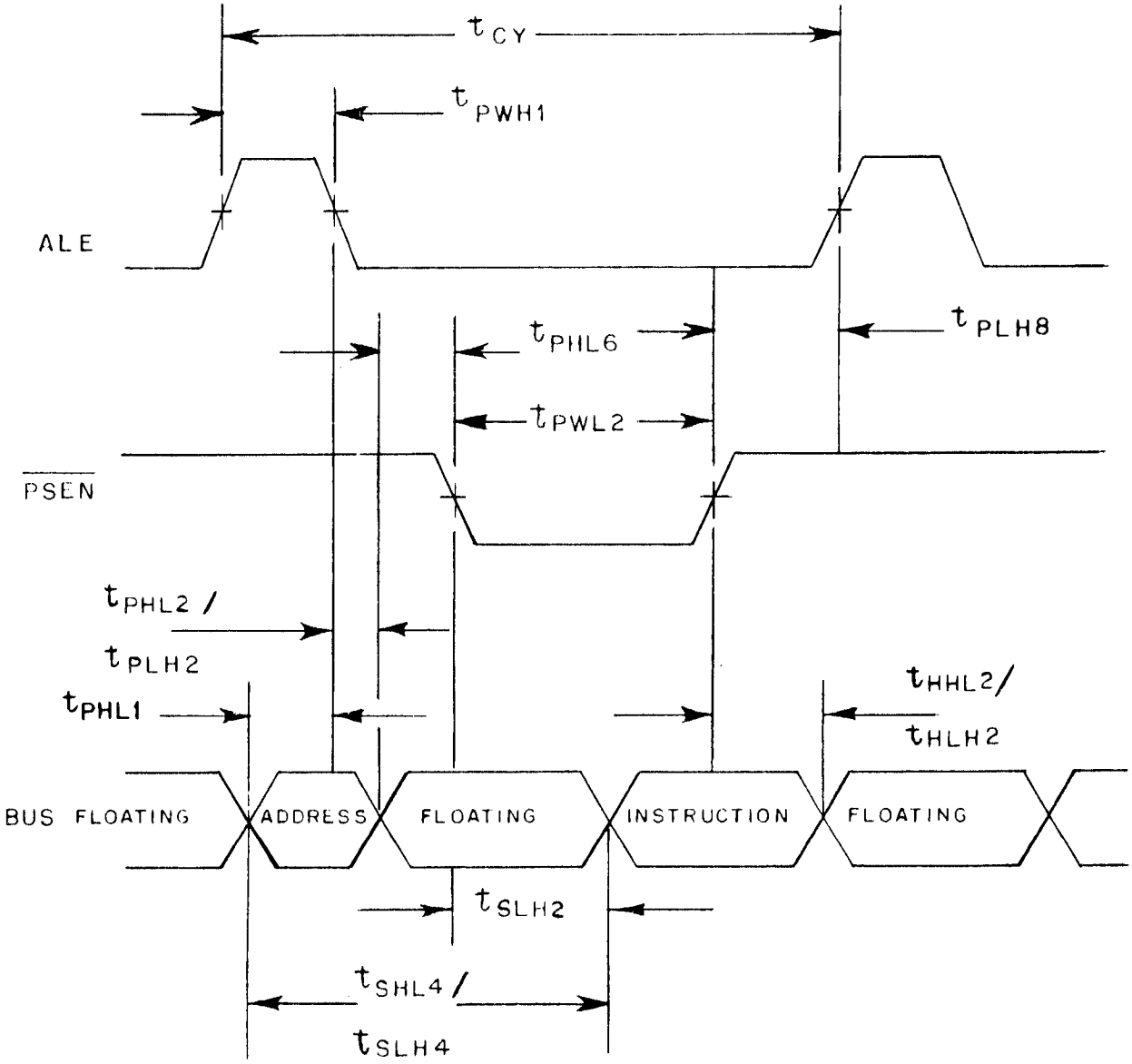


FIGURE 6. Instruction fetch from external program memory.

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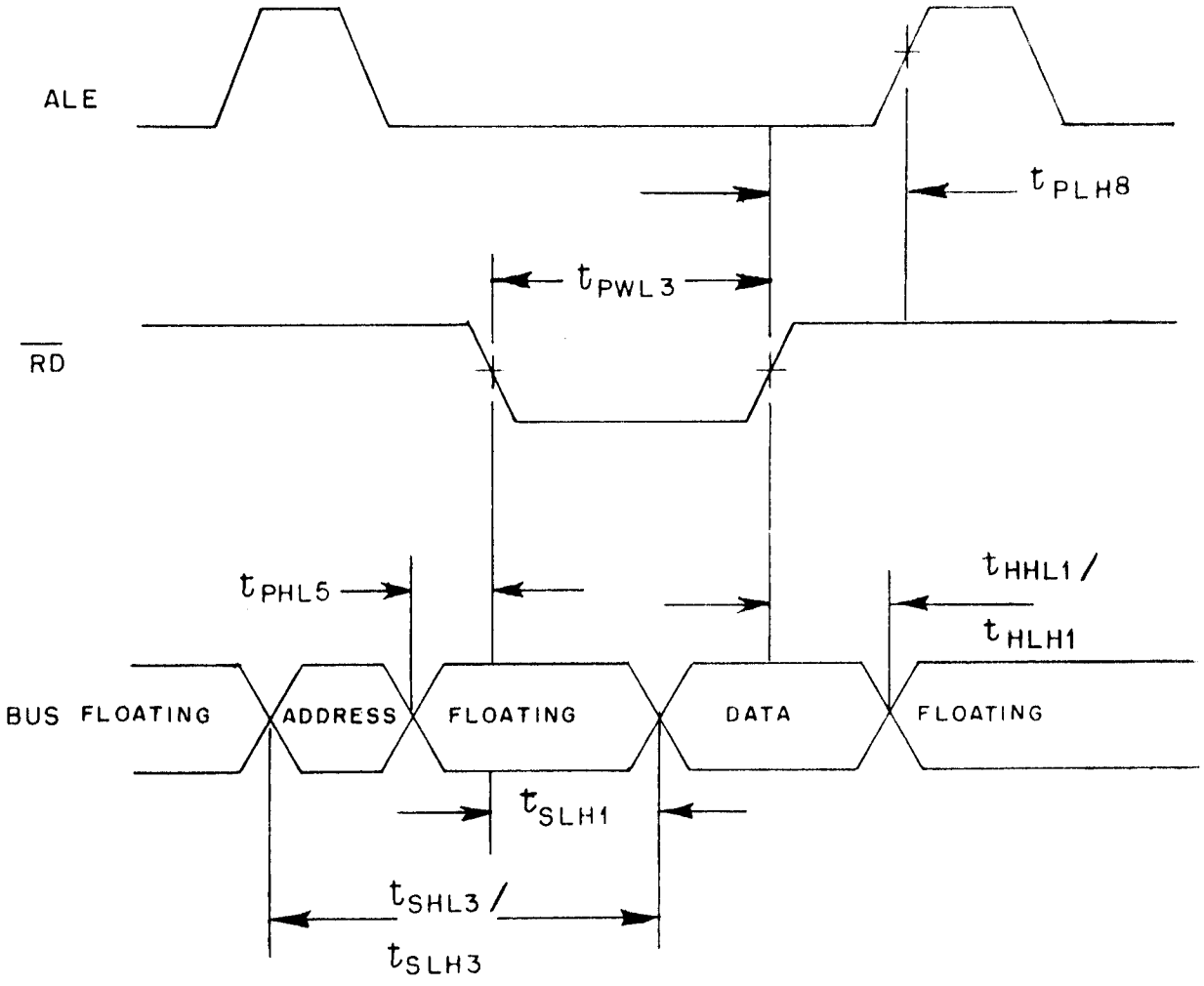


FIGURE 7. Read from external data memory.

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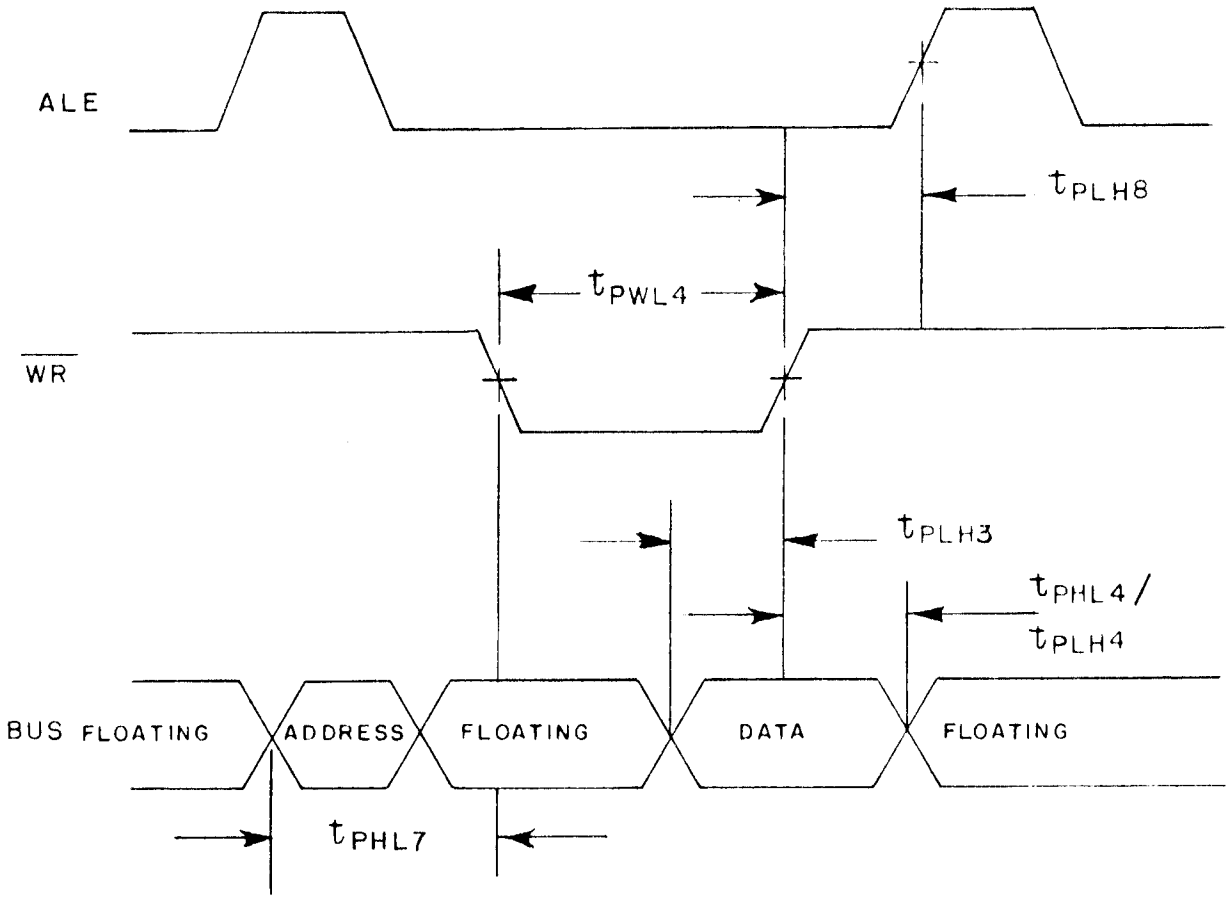


FIGURE 8. Write to external data memory.

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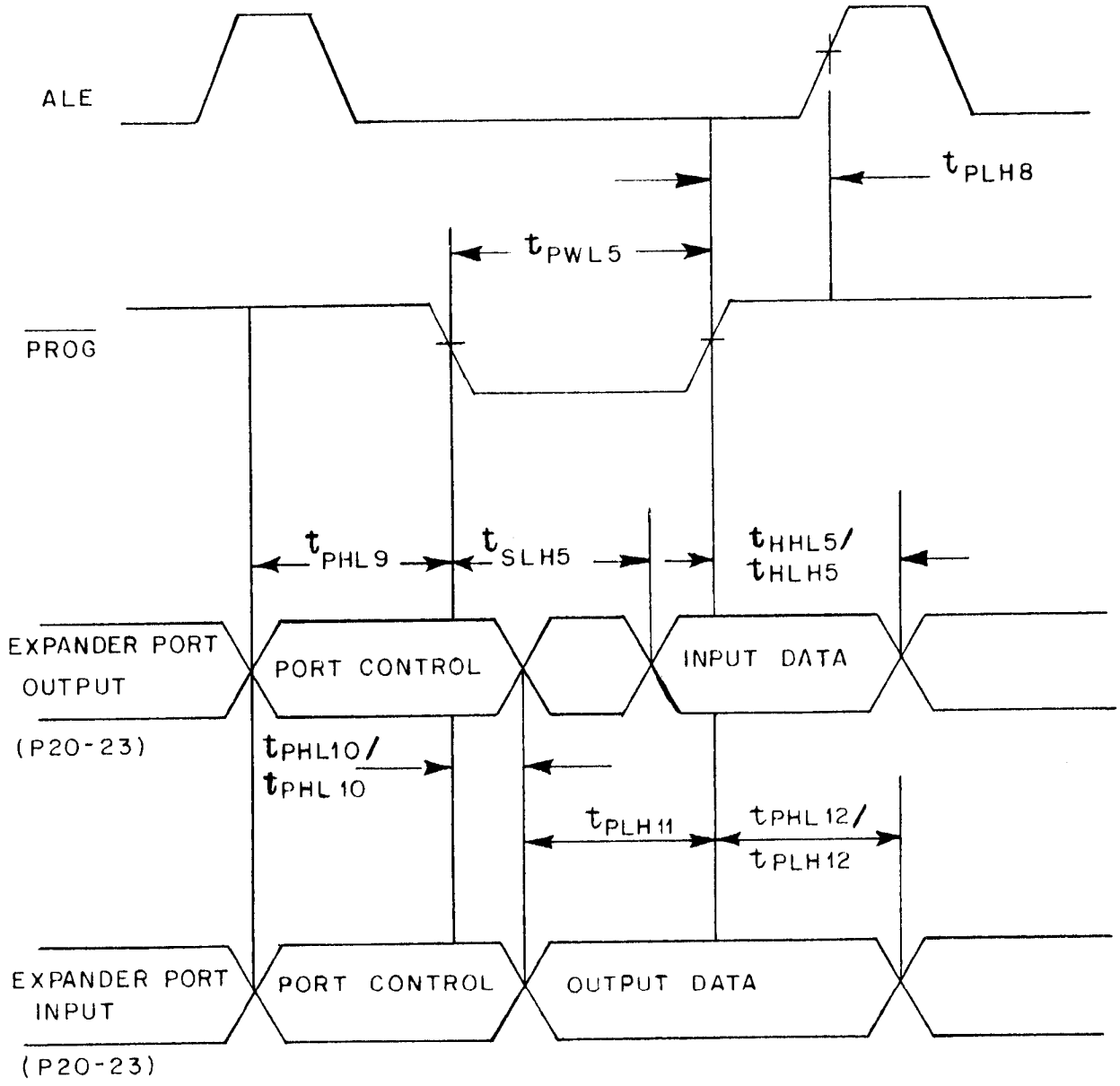
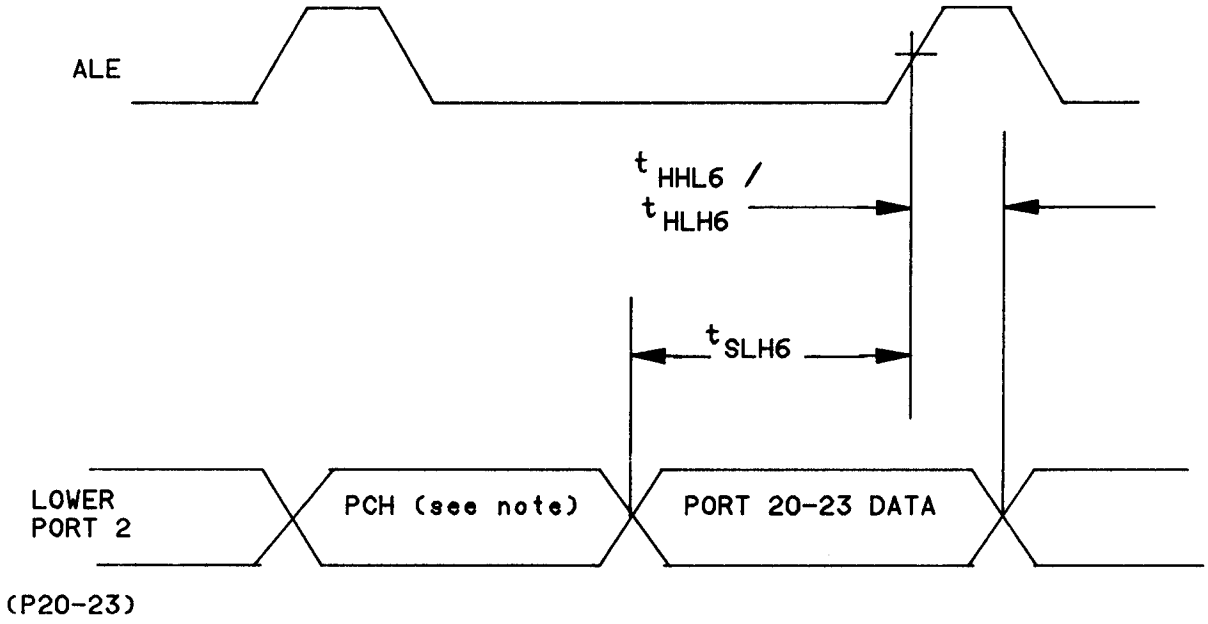


FIGURE 9. Lower port 2 used as an expander port.

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NOTE: The program counter high order bits are outputted on LOWER PORT 2 during this time.

FIGURE 10. Lower port 2 when used for normal I/O data.

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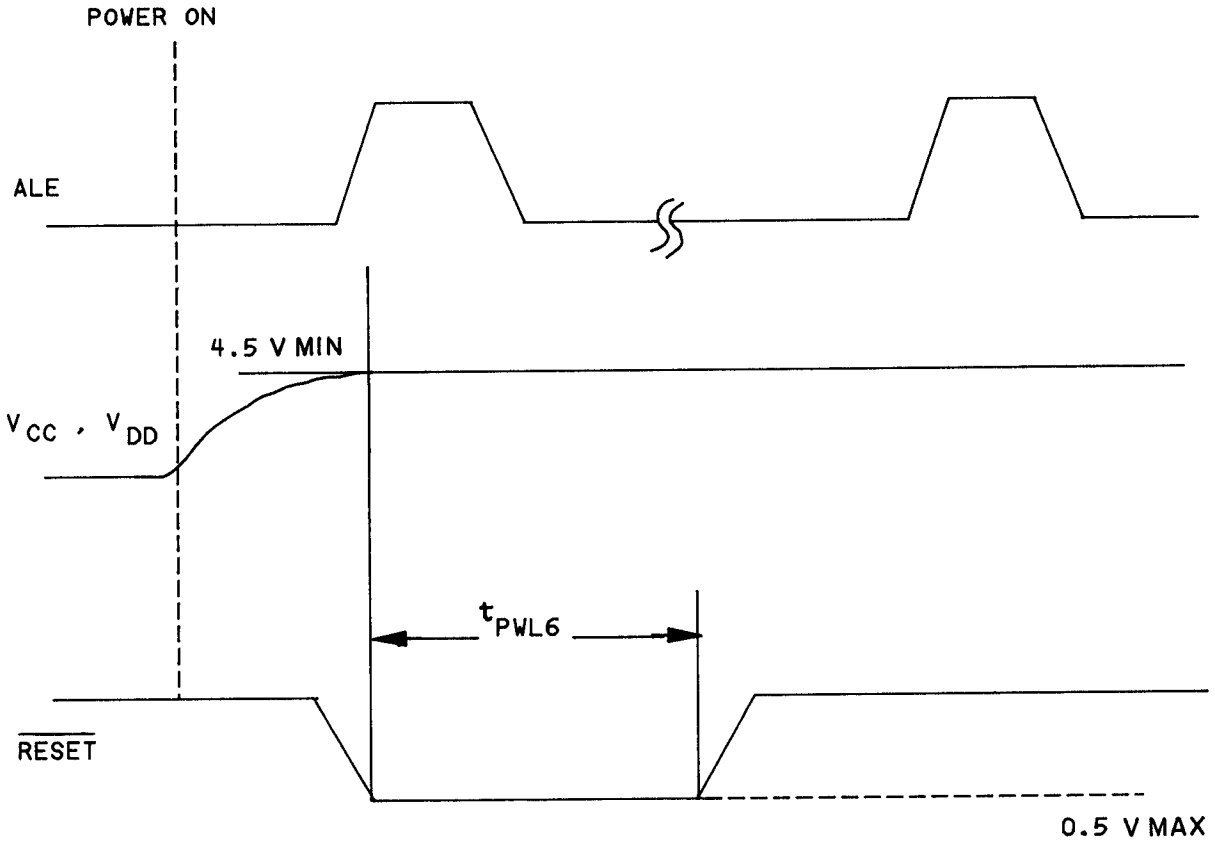
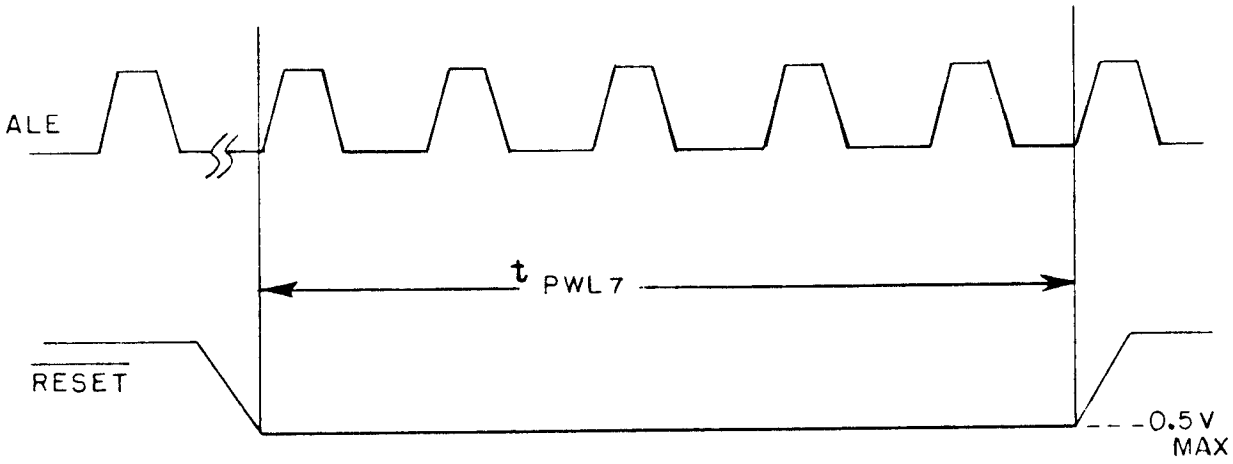


FIGURE 11. $\overline{RESET}$ requirements during initialization.

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NOTES:

1. The oscillator must be stabilized before the $\overline{\text{reset}}$ pulse is applied.
2. $\overline{\text{Reset}}$ must be held low for only 5 machine cycles (12.5 μs if the crystal input frequency is 6 MHz).

FIGURE 12. $\overline{\text{Reset}}$ pulse requirements after the oscillator has stabilized.

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TABLE IIIA. Input conditions for table IIIC, device types 01 and 02.

Symbol	Waveform figure	Terminals	Input conditions						Unit
			A	B	C	D	E	F	
V_{CC}, V_{DD}	---	V_{CC}, V_{DD}	5.5	4.5	5.5	4.5	4.5	4.5	V
V_{IH} 1/	---	$\overline{RESET}$ XTAL1 XTAL2	3.8	3.8	3.8	3.8	3.8	3.8	V
V_{IH}	---	All other inputs	2.2	2.2	2.2	2.2	2.2	2.2	V
V_{IL} 1/	---	$\overline{RESET}$ XTAL1 XTAL2	0.5	0.5	0.5	0.5	0.5	0.5	V
V_{IL}	---	All other inputs	0.7	0.7	0.7	0.7	0.7	0.7	V
Crystal input frequency	---	XTAL1 XTAL2	6.0 X1 High-35% X2 High-65%	6.0 X1 High-35% X2 High-65%	6.0 X1 High-65% X2 High-35%	6.0 X1 High-65% X2 High-35%	1 X1 High-35% X2 High-65%	1 X1 High-35% X2 High-65%	MHz
t_{SLH1}	11	BUS	280	280	280	280	4900	4900	ns
t_{SLH2}	10	BUS	400	400	400	400	3400	3400	ns

See footnote at end of table.

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TABLE IIIA. Input conditions for table IIIC, device types 01 and 02 - Continued.

Symbol	Waveform figure	Terminals	Input conditions						Unit
			A	B	C	D	E	F	
t_{SHL3} t_{SLH3}	7	BUS	1330	1330	1330	1330	10,350	10,350	ns
t_{SHL4} t_{SLH4}	6	BUS	830	830	830	830	7850	7850	ns
t_{SLH5}	9	Expander Port (P20-23)	1020	1020	1020	1020	8790	8790	ns
t_{SLH6}	10	Port 2	360	360	360	360	3200	3200	ns
t_{HHL1} t_{HLH1}	7	BUS	0	0	0	0	0	0	ns
t_{HHL2} t_{HLH2}	6	BUS	0	0	0	0	0	0	ns
t_{HHL5} t_{HLH5}	9	Expander Port (P20-23)	0	0	0	0	0	0	ns
t_{HHL6} t_{HLH6}	10	Port 2	50	50	50	50	800	800	ns
t_{PWL7} 1/	12	$\overline{RESET}$	12.5	12.5	12.5	12.5	75	75	µs

1/ The $\overline{RESET}$ input must be held low (see figures 11 and 12) prior to making I_{IC} measurements on ports 1 and 2.

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TABLE IIIB. Output conditions for table IIIC, device types 01 and 02.

Symbol	Waveform figure	Terminals	Input conditions		Unit
			A	B	
V_{OH}	---	All outputs	2.4	2.4	V
V_{OL}	---	All outputs	0.45	0.45	V
t_{PHL1}	6	BUS	290	2290	ns
t_{PHL2} t_{PLH2}	6	BUS	20	360	ns
t_{PLH3}	8	BUS	750	5700	ns
t_{PHL4} t_{PLH4}	8	BUS	10	350	ns
t_{PHL5}	7	BUS	190	1360	ns
t_{PHL6}	6	BUS	-40	460	ns
t_{PHL7}	8	BUS	750	5250	ns
t_{PLH8}	6, 7, 8, 9	$\overline{PSEN}$, $\overline{WR}$, $\overline{RD}$, and $\overline{PROG}$	10	10	ns
t_{PHL9}	9	Expander Port (P20-23)	120	1320	ns
t_{PHL10} t_{PLH10}	9	Expander Port (P20-23)	360	3200	ns
t_{PLH11}	9	Expander Port (P20-23)	700	5250	ns
t_{PHL12} t_{PLH12}	9	Expander Port (P20-23)	100	850	ns
t_{cy}	6	ALE	2.5	15	µs
t_{PWH1}	6	ALE	280	3,730	ns

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TABLE IIIB. Output conditions for table IIIC, device types 01 and 02 - Continued.

Symbol	Waveform figure	Terminals	Input conditions		Unit
			A	B	
t_{PWL2}	6	$\overline{PSEN}$	600	5200	ns
t_{PWL3}	7	$\overline{RD}$	950	6700	ns
t_{PWL4}	8	$\overline{WR}$	950	6700	ns
t_{PWL5}	9	$\overline{PROG}$	1250	9650	ns
t_{PWH8}	---	T0	56	336	ns
t_{CLK}	---	T0	0.5	3.0	µs

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TABLE IIIC. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				T0	XTAL1	XTAL2	$\overline{RESET}$	$\overline{SS}$	$\overline{INT}$	EA	$\overline{RD}$	$\overline{PSEN}$	$\overline{WR}$	ALE	DB0	DB1	DB2	DB3	DB4	DB5	DB6	DB7	V_{SS}
1 TC = 25°C	I_{IH1}		1	5.5 V			GND		5.5 V						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND
	"		2				GND																"
	"		3				"																"
	"		4				"																"
	"		5				"																"
	"		6				"																"
	"		7				"																"
	"		8				"																"
	"		9				"																"
	"		10				"																"
	"		11				"																"
	I_{IH2}	1/	12					5.5 V															"
	"	"	13					"															"
	"	"	14					"															"
	"	"	15					"															"
	"	"	16					"															"
	"	"	17					"															"
	"	"	18					"															"
	"	"	19					"															"
	"	"	20					"															"
	"	"	21					"															"
	"	"	22					"															"
	"	"	23					"															"
	"	"	24					"															"
	"	"	25					"															"
	"	"	26					"															"
	"	"	27					"															"
	"	"	28					"															"
	"	"	29					"															"
	I_{IH3}		30		5.5 V																		"
	I_{IH4}		31				5.5 V																"
	I_{IH5}		32																				"
	I_{IL1}		33	0.45 V			GND																"
	"		34				GND		0.45 V														"
	"		35				"								0.45 V								"
	"		36				"									0.45 V							"
	"		37				"										0.45 V						"
	"		38				"											0.45 V					"
	"		39				"												0.45 V				"
	"		40				"													0.45 V			"
	"		41				"														0.45 V		"
	"		42				"															0.45 V	"
	"		43				"																"
	I_{IL2}	1/	44					0.45 V															"
	"	"	45					"		0.45 V													"
	"	"	46					"															"
	"	"	47					"															"
	"	"	48					"															"
	"	"	49					"															"
	"	"	50					"															"
	"	"	51					"															"
	"	"	52					"															"
	"	"	53					"															"
	"	"	54					"															"
	"	"	55					"															"
	"	"	56					"															"
	"	"	57					"															"
	"	"	58					"															"
	"	"	59					"															"
	"	"	60					"															"
	"	"	61					"															"

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	P20	P21	P22	P23	$\overline{PROG}$	V_{DD}	P10	P11	P12	P13	P14	P15	P16	P17	P24	P25	P26	P27	T1	V_{CC}		Min	Max	
1						5.5 V														5.5 V	1	-10	10	µA
2						"														"	6	"	"	"
3						"														"	12	"	"	"
4						"														"	13	"	"	"
5						"														"	14	"	"	"
6						"														"	15	"	"	"
7						"														"	16	"	"	"
8						"														"	17	"	"	"
9						"														"	18	"	"	"
10						"														"	19	"	"	"
11						"													5.5 V	"	39	"	"	"
12						"														"	5		-700	"
13						"														"	7		"	"
14	5.5 V					"														"	21		"	"
15		5.5 V				"														"	22		"	"
16			5.5 V			"														"	23		"	"
17				5.5 V		"														"	24		"	"
18						"	5.5 V													"	27		"	"
19						"		5.5 V												"	28		"	"
20						"			5.5 V											"	29		"	"
21						"				5.5 V										"	30		"	"
22						"					5.5 V									"	31		"	"
23						"						5.5 V								"	32		"	"
24						"							5.5 V							"	33		"	"
25						"								5.5 V						"	34		"	"
26						"									5.5 V					"	35		"	"
27						"										5.5 V				"	36		"	"
28						"											5.5 V			"	37		"	"
29						"												5.5 V		"	38		"	"
30						"														"	2	-50	50	"
31						"														"	4		-700	"
32						"														"	3		13	mA
33						"														"	1	-10	10	µA
34						"														"	6	"	"	"
35						"														"	12	"	"	"
36						"														"	13	"	"	"
37						"														"	14	"	"	"
38						"														"	15	"	"	"
39						"														"	16	"	"	"
40						"														"	17	"	"	"
41						"														"	18	"	"	"
42						"														"	19	"	"	"
43						"													0.45 V	"	39	"	"	"
44						"														"	5		-700	"
45						"														"	7		"	"
46	0.45 V					"														"	21		"	"
47		0.45 V				"														"	22		"	"
48			0.45 V			"														"	23		"	"
49				0.45 V		"														"	24		"	"
50						"	0.45 V													"	27		"	"
51						"		0.45 V												"	28		"	"
52						"			0.45 V											"	29		"	"
53						"				0.45 V										"	30		"	"
54						"					-.45 V									"	31		"	"
55						"						0.45 V								"	32		"	"
56						"							0.45 V							"	33		"	"
57						"								0.45 V						"	34		"	"
58						"									04.5 V					"	35		"	"
59						"										0.45 V				"	36		"	"
60						"											0.45 V			"	37		"	"
61						"												0.45 V		"	38		"	"

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				T0	XTAL1	XTAL2	RESET	SS	INT	EA	RD	PSEN	WR	ALE	DB0	DB1	DB2	DB3	DB4	DB5	DB6	DB7	V_{SS}
1 T_C = 25°C	I_{IL3}		62		0.45 V																		GND
	I_{IL4}		63				0.45 V																"
	I_{IL5}		64			0.45 V																	"
	I_{DD1}		65																				"
	I_{DD2}		66																				"
	I_{CC}		67																				"
	V_{IC}		68	-200 µA																			"
	"		69		-200 µA																			"
	"		70			-200 µA																		"
	"		71				-200 µA																	"
	"		72					-200 µA																"
	"		73						-200 µA															"
	"		74							-200 µA														"
	"		75												-200 µA									"
	"		76													-200 µA								"
	"		77														-200 µA							"
	"		78															-200 µA						"
	"		79																-200 µA					"
	"		80																	-200 µA				"
	"		81																		-200 µA			"
	"		82																			-200 µA		"
	"		83																					"
	"		84																					"
	"		85																					"
	"		86																					"
	"		87																					"
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	"		98																					"
	"		99																					"

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	P20	P21	P22	P23	$\overline{PROG}$	V_{DD}	P10	P11	P12	P13	P14	P15	P16	P17	P24	P25	P26	P27	T1	V_{CC}		Min	Max	
62						5.5 V														5.5 V	2	-50	50	µA
63						"														"	4		-700	µA
64						"														"	3		13	mA
65						"														"	26		10	"
66						"														GND	26		10	"
67						"														5.5 V	40		90	"
68						"														"	1		-4.0	V
69						"														"	2		"	"
70						"														"	3		"	"
71						"														"	4		"	"
72						"														"	5		"	"
73						"														"	6		"	"
74						"														"	7		"	"
75						"														"	12		"	"
76						"														"	13		"	"
77						"														"	14		"	"
78						"														"	15		"	"
79						"														"	16		"	"
80						"														"	17		"	"
81						"														"	18		"	"
82						"														"	19		"	"
83	-200 µA					"														"	21		"	"
84		-200 µA				"														"	22		"	"
85			-200 µA			"														"	23		"	"
86				-200 µA		"														"	24		"	"
87						"	-200 µA													"	27		"	"
88						"		-200 µA												"	28		"	"
89						"			-200 µA											"	29		"	"
90						"				-200 µA										"	30		"	"
91						"					-200 µA									"	31		"	"
92						"						-200 µA								"	32		"	"
93						"							-200 µA							"	33		"	"
94						"								-200 µA						"	34		"	"
95						"									-200 µA					"	35		"	"
96						"										-200 µA				"	36		"	"
97						"											-200 µA			"	37		"	"
98						"												-200 µA		"	38		"	"
99						"													-200 µA	"	39		"	"

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device type 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3		5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				TO	XTAL1	XTAL2	$\overline{RESET}$	$\overline{SS}$	$\overline{INT}$	EA	$\overline{RD}$	$\overline{PSEN}$	$\overline{WR}$	ALE	DB0	DB1	DB2	DB3	DB4	DB5	DB6	DB7	V_{SS}
4 T_C = 25°C	C_I	3012	100	2/	2/	2/	2/	2/	2/	2/					2/	2/	2/	2/	2/	2/	2/	2/	GND
	"	"	101																				"
	"	"	102																				"
	"	"	103																				"
	"	"	104																				"
	"	"	105																				"
	"	"	106																				"
	"	"	107																				"
	"	"	108																				"
	"	"	109																				"
	"	"	110																				"
	"	"	111																				"
	"	"	112																				"
	"	"	113																				"
	"	"	114																				"
	"	"	115																				"
	"	"	116																				"
	"	"	117																				"
	"	"	118																				"
	"	"	119																				"
	"	"	120																				"
	"	"	121																				"
	"	"	122																				"
	"	"	123																				"
	"	"	124																				"
	"	"	125																				"
	"	"	126																				"
	"	"	127																				"
	"	"	128																				"
	"	"	129																				"
	"	"	130																				"
	"	"	131																				"
	C_{O_4}	"	132								2/												"
	"	"	133									2/											"
	"	"	134										2/										"
	"	"	135											2/									"
	"	"	136																				"
2 T_C = 125°C	Repeat tests 1 through 67 at 125°C.																						
3 T_C = -55°C	Repeat tests 1 through 67 at -55°C.																						
7 T_C = 25°C 3/	Dynamic Functional test	MIL-STD-883 methods 3006, 3007, and 3014	Table IV - Input condition A of table IIIA and output condition A of table IIIB.																				
			Table IV - Input condition B of table IIIA and output condition A of table IIIB.																				
			Table IV - Input condition C of table IIIA and output condition A of table IIIB.																				
			Table IV - Input condition D of table IIIA and output condition A of table IIIB.																				
			Table IV - Input condition E of table IIIA and output condition B of table IIIB.																				
			Table IV - Input condition F of table IIIA and output condition B of table IIIB.																				
7 T_C = 25°C 4/	Verification of ROM contents (Device type 01 only)	The internal program memory contents shall be read out and verified using the procedure described in 5/ of the table III notes with $V_{CC} = V_{DD} = 4.5$ V.																					
		Repeat above test with $V_{CC} = V_{DD} = 5.5$ V																					
8 T_C=+125°C	Repeat subgroup 7 at +125°C. 4/																						
T_C=-55°C	Repeat subgroup 7 at -55°C. 4/																						

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits Min	Test limits Max	Unit
	P20	P21	P22	P23	$\overline{PROG}$	V_{DD}	P10	P11	P12	P13	P14	P15	P16	P17	P24	P25	P26	P27	T1	V_{CC}				
100						5.5 V														5.5 V	1	2	30	pF
101						"														"	2	"	"	"
102						"														"	3	"	"	"
103						"														"	4	"	"	"
104						"														"	5	"	"	"
105						"														"	6	"	"	"
106						"														"	7	"	"	"
107						"														"	12	"	"	"
108						"														"	13	"	"	"
109						"														"	14	"	"	"
110						"														"	15	"	"	"
111						"														"	16	"	"	"
112						"														"	17	"	"	"
113						"														"	18	"	"	"
114						"														"	19	"	"	"
115	2/					"														"	21	"	"	"
116		2/				"														"	22	"	"	"
117			2/			"														"	23	"	"	"
118				2/		"														"	24	"	"	"
119						"	2/													"	27	"	"	"
120						"		2/												"	28	"	"	"
121						"			2/											"	29	"	"	"
122						"				2/										"	30	"	"	"
123						"					2/									"	31	"	"	"
124						"						2/								"	32	"	"	"
125						"							2/							"	33	"	"	"
126						"								2/						"	34	"	"	"
127						"									2/					"	35	"	"	"
128						"										2/				"	36	"	"	"
129						"											2/			"	37	"	"	"
130						"												2/		"	38	"	"	"
131						"													2/	"	39	"	"	"
132						"														"	8	"	20	"
133						"														"	9	"	"	"
134						"														"	10	"	"	"
135						"														"	11	"	"	"
136					2/	"														"	25	"	"	"

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Symbol		Test no.	Test pattern number	Waveform figure	Measurement terminals: Dip: Input under test	Measurement terminals: Dip: Output under test	Measurement terminals: Flatpack: Input under test	Measurement terminals: Flatpack: Output under test	Test limits: Subgroup 9 T_C = +25°C: Min	Test limits: Subgroup 9 T_C = +25°C: Max	Test limits: Subgroup 10 T_C = 125°C: Min	Test limits: Subgroup 10 T_C = 125°C: Max	Test limits: Subgroup 11 T_C = -55°C: Min	Test limits: Subgroup 11 T_C = -55°C: Max	Unit
Crystal input tests	1 MHz			Monitor ALE output for correct pulses per table I with a 1 MHz crystal connected to XTAL1 and XTAL2 and $V_{CC} = V_{DD} = 4.5$ V.											
				Repeat above test with $V_{CC} = V_{DD} = 5.5$ V.											
	6 MHz			Monitor ALE output for correct pulses per table I with a 6 MHz crystal connected to XTAL1 and XTAL2 and $V_{CC} = V_{DD} = 4.5$ V.											
				Repeat above test with $V_{CC} = V_{DD} = 5.5$ V.											
				Repeat above tests for 1 MHz and 6 MHz and $T_C = 125°C$.											
				Repeat above tests for 1 MHz and 6 MHz and $T_C = -55°C$.											

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.

Symbol	Test no.	Test pattern number	Waveform figure	Terminal(s)	MIL-STD-883 method	Subgroup 12 limits +25°C, +125°C				Unit
						Crystal input frequency				
						6 MHz		1 MHz		
						Min	Max	Min	Max	
t_{PWH1}	137		6	ALE	3003	280		1680		ns
t_{PWL2}	138		6	$\overline{PSEN}$	"	600		3600		ns
t_{PWL3}	139		7	$\overline{RD}$	"	950		2400		ns
t_{PWL4}	140		8	$\overline{WR}$	"	950		2400		ns
t_{PWL5}	141		9	$\overline{PROG}$	"	1250		7500		ns
t_{PWH8}	142		---	T0	"	56		336		ns
t_{CLK}	143		---	T0	"	0.5		3.0		µs
t_{cy}	144		6	ALE	"	2.5		15.0		µs
t_{PHL1}	145		6	BUS	"	290		290		ns
t_{PHL2} 6/	146		6	BUS	"	20		20		ns
t_{PLH2}			6	BUS	"	20		20		ns
t_{PLH3}	147		8	BUS	"	730		730		ns
t_{PHL4} 6/	148		8	BUS	"	10		10		ns
t_{PLH4}			8	BUS	"	10		10		ns
t_{PHL5}	149		7	$\overline{RD}$	"	190		190		ns
t_{PHL6}	150		6	$\overline{PSEN}$	"	-40		-40		ns
t_{PHL7}	151		8	$\overline{WR}$	"	500		500		ns
t_{PLH8}	152		6,7,8,9	ALE	"	10		10		ns
t_{PHL9}	153		9	$\overline{PROG}$	"	120		120		ns
t_{PHL10} 6/	154		9	P20-23	"	360		360		ns
t_{PLH10}			9	P20-23	"	360		360		ns
t_{PLH11}	155		9	$\overline{PROG}$	"	700		700		ns
t_{PHL12} 6/	156		9	P20-23	"	100		100		ns
t_{PLH12}			9	P20-23	"	100		100		ns

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.

Symbol	Test no.	Test pattern number	Waveform figure	Terminal	MIL-STD-883 method	Subgroup 12 limits +25°C, +125°C Crystal input frequency 6 MHz Min	6 MHz Max	1 MHz Min	1 MHz Max	Unit
V_{OH1}	157			BUS	3006	2.4		2.4		V
V_{OH2}	158			ALE	"	2.4		2.4		V
V_{OH2}	159			$\overline{PSEN}$	"	2.4		2.4		V
V_{OH2}	160			$\overline{RD}$	"	2.4		2.4		V
V_{OH2}	161			$\overline{WR}$	"	2.4		2.4		V
V_{OH3}	162			PORT 1	"	2.4		2.4		V
V_{OH3}	163			PORT 2	"	2.4		2.4		V
V_{OH3}	164			$\overline{PROG}$	"	2.4		2.4		V
V_{OH3}	165			TO	"	2.4		2.4		V
V_{OL1}	166			BUS	3007		0.45		0.45	V
V_{OL2}	167			PORT 1	3007		0.45		0.45	V
V_{OL2}	168			PORT 2	3007		0.45		0.45	V

1/ The $\overline{RESET}$ input must be held low (see figures 11 and 12) prior to making I_{IH} and I_{IL} measurements on ports 1 and 2.

2/ The capacitance measurements shall be made between the indicated terminal and the grounded terminal(s) at a frequency of 1 MHz. The dc bias of the measuring instrument shall be less than ±0.1 V. The ac signal amplitude shall be less than 50 mV rms.

3/ When the vectors are performed as specified in table III for subgroups 7 and 8, they verify the dynamic, functional, output level, threshold, and switching parameters at 25°C, -55°C, and +125°C.

4/ For subgroups 7 and 8 XTAL1 and XTAL2 inputs are driven by the tester per table IIIA.

5/ In order to place device type 01 in the Read mode to verify the internal program memory, the following procedure must be performed:

a. Apply +12 V to the EA pin ($\overline{RESET}$ must be at 0 V when voltage is applied to EA).

b. The address of the location to be read is then applied to the BUS and lower port 2 lines. The address is latched by a "0" to "1" transition on RESET and a high level on RESET causes the contents of the program memory location addressed to appear on the 8 lines of BUS.

c. $\overline{RESET}$ must be brought back to 0 V before leaving the READ mode.

6/ Where propagation delay tests for subgroup 12 indicate a group of pins and two or more propagation delays are specified for a test, only the readings for the pin(s) with the worst case minimum and maximum delay require recording. Variables data shall be taken for subgroup 12 only during initial qualification and after process or design changes.

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TABLE IV. Instruction set.

Accumulator

Mnemonic	Description	Bytes	Cycles
ADD A, R	Add register to A	1	1
ADD A, @R	Add data memory to A	1	1
ADD A, # data	Add immediate to A	2	2
ADDC A, R	Add register with carry	1	1
ADDC A, @R	Add data memory with carry	1	1
ADDC A, # data	Add immediate with carry	2	2
ANL A, R	And register to A	1	1
ANL A, @R	And data memory to A	1	1
ANL A, # data	And immediate to A	2	2
ORL A, R	Or register to A	1	1
ORL A @R	Or data memory to A	1	1
ORL A, # data	Or immediate to A	2	2
XRL A, R	Exclusive or register to A	1	1
XRL A, @R	Exclusive or data memory to A	1	1
XRL, A, # data	Exclusive or immediate to A	2	2
INC A	Increment A	1	1
DEC A	Decrement A	1	1
CLR A	Clear A	1	1
CPL A	Complement A	1	1
DA A	Decimal adjust A	1	1
SWAP A	Swap nibbles of A	1	1
RL A	Rotate A left	1	1
RLC A	Rotate A left through carry	1	1
RR A	Rotate A right	1	1
RRC A	Rotate A right through carry	1	1

Input/Output

Mnemonic	Description	Bytes	Cycles
IN A, P	Input port to A	1	2
OUTL P, A	Output A to port	1	2
ANL P, # data	And immediate to port	2	2
ORL P, # data	Or immediate to port	2	2
INS A, BUS	Input BUS to A	1	2
OUTL BUS, A	Output A to BUS	1	2
ANL BUS, # data	And immediate to BUS	2	2
ORL BUS, # data	Or immediate to BUS	2	2
MOVD A,P	Input expander port to A	1	2
MOVD P, A	Output A to expander port	1	2
ANLD P, A	And A to expander port	1	2
ORLD P, A	Or A to expander port	1	2

Registers

Mnemonic	Description	Bytes	Cycles
INC R	Increment register	1	1
INC @R	Increment data memory	1	1
DEC R	Decrement register	1	1

Branch

Mnemonic	Description	Bytes	Cycles
JMP addr	Jump unconditional	2	2
JMPP @A	Jump indirect	1	2
DJNZ R, addr	Decrement register and skip	2	2
JC addr	Jump on carry = 1	2	2
JNC addr	Jump on carry = 0	2	2
JZ addr	Jump on A zero	2	2
JNZ addr	Jump on A not zero	2	2
JTO addr	Jump on TO = 1	2	2
JNTO addr	Jump on TO = 0	2	2
JT1 addr	Jump on T1 = 1	2	2
JNT1 addr	Jump on T1 = 0	2	2
JF0 addr	Jump on F0 = 1	2	2
JF1 addr	Jump on F1 = 1	2	2
JTF addr	Jump on timer flag	2	2
JN1 addr	Jump on INT = 0	2	2
JBb addr	Jump on accumulator bit	2	2

Subroutine

Mnemonic	Description	Bytes	Cycles
CALL addr	Jump to subroutine	2	2
RETR	Return	1	2
RETR	Return and restore status	1	2

Flags

Mnemonic	Description	Bytes	Cycles
CLR C	Clear carry	1	1
CPL C	Complement carry	1	1
CLR F0	CLear flag 0	1	1
CPL F0	Complement flag 0	1	1
CLR F1	Clear flag 1	1	1
CPL F1	Complement flag 1	1	1

Data Moves

Mnemonic	Description	Bytes	Cycles
MOV A, R	Move register to A	1	1
MOV A, @R	Move data memory to A	1	1
MOV A, # data	Move immediate to A	2	2
MOV R, A	Move A to register	1	1
MOV @R, A	Move A to data memory	1	1
MOV R, # data	Move immediate to register	2	2
MOV @R, #data	Move immediate to data memory	2	2
MOV A, PSW	Move PSW to A	1	1
MOV PSW, A	Move A to PSW	1	1
XCH A, R	Exchange A and register	1	1
XCH A, @R	Exchange A and data memory	1	1
XCHD A, @R	Exchange nibble of A and register	1	
MOVX A, @R	Move external data memory to A	1	2
MOVX @R, A	Move A to external data memory	1	2
MOVP A, @A	Move to A from current page	1	2
MOVP3 A, @	Move to A from page 3	1	2

Timer/Counter

Mnemonic	Description	Bytes	Cycles
MOV A, T	Read timer/counter	1	1
MOV T, A	Load timer/counter	1	1
STRT T	Start timer	1	1
STRT CNT	Start counter	1	1
STOP TCNT	Stop timer/counter	1	1
EN TCNTI	Enable timer/counter interrupt	1	1
DIS TCNTI	Disable timer/counter interrupt	1	1

Control

Mnemonic	Description	Bytes	Cycles
EN I	Enable external interrupt	1	1
DIS I	Disable external interrupt	1	1
SEL RB0	Select register bank 0	1	1
SEL RB1	Select register bank 1	1	1
SEL MB0	Select memory bank 0	1	1
SEL MB1	Select memory bank 1	1	1
ENT 0 CLK	Enable clock output on T0	1	1

Mnemonic	Description	Bytes	Cycles
NOP	No operation	1	1

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4.4.3 Groups C and D inspections. Groups C and D inspections shall be in accordance with tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II.

b. Steady-state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) Temperature (T_A) shall be 125°C.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 volts from a 100 pF source in the following test sequence:

a. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 2 pulse sequence as follows and in accordance with the test circuit of figure 3.

(1) Input (-) to V_{SS}.

(2) Input (+) to V_{SS}.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. The devices covered by this specification require electrostatic protection.

6. NOTES

6.1 Computer generated table. Table V, which forms a part of this specification is not printed herein because of its extreme length; and even when it is presented as hard copy it requires conversion to on-line format.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

PIN DESCRIPTION

Designation	Pin no.	Function
V_{SS}	20	Ground zero voltage potential.
V_{DD}	26	RAM power supply; +5 V during normal operation and standby.
V_{CC}	40	Main power supply; +5 V during normal operation.
$\overline{PROG}$	25	Output strobe for 8243 I/O expander.
P10-P17 Port 1	27-34	8-bit quasi-bidirectional port.
P20-P27 Port 2	21-24 35-38	8-bit quasi-bidirectional port. P20-P23 contain the four high order program counter bits during an external program memory fetch and serve as a 4-bit I/O expander bus for 8243.
DB_0-DB_7 BUS	12-19	True bidirectional port which can be written or read synchronously using the $\overline{RD}$, $\overline{WR}$ strobes. The port can also be statically latched. Contains the 8 low order program counter bits during an external program memory fetch, and receives the addressed instruction under the control of PSEN. Also contains the address and data during an external RAM data store instruction, under control of ALE, RD, and WR.
T0	1	Input pin testable using the conditional transfer instructions JT0 and JNT0. T0 can be designated as a clock output using ENT0 CLK instruction.

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T1	39	Input pin testable using the JT1, and JNT1 instructions. Can be designated the timer/counter input using the STRT CNT instruction.
$\overline{INT}$	6	Interrupt input. Initiates an interrupt if interrupt is enabled. Interrupt is disabled after a reset. Also testable with conditional jump instruction. (Active low)
$\overline{RD}$	8	Output strobe activated during a BUS read. Can be used to enable data onto the BUS from an external device. Used as a read strobe to external data memory. (Active low)
$\overline{RESET}$	4	Input which is used to initialize the processor. Also used during power down. (Active low) (Non TTL V_{IH}) Performs the following functions: 1. Sets program counter to zero. 2. Sets stack pointer to zero. 3. Selects register bank 0. 4. Selects memory bank 0. 5. Sets BUS to high impedance state. (Except when EA = 5 V) 6. Sets ports 1 and 2 to input mode. 7. Disables interrupts (timer and external). 8. Stops timer. 9. Clears timer flag. 10. Clears F0 and F1. 11. Disables clock output from T0.
$\overline{WR}$	10	Output strobe during a BUS write. (Active low) Used as write strobe to external data memory
ALE	11	Address latch enable. This signal occurs once during each cycle and is useful as a clock output. The negative edge of ALE strobes address into external data and program memory.
$\overline{PSEN}$	9	Program strobe enable. This output occurs only during a fetch to external program memory. (Active low)
EA	7	External access input which forces all program memory fetches to reference external memory. Useful for emulation and debug, and essential for testing and program verification. (Active high.) The EA pin must be held at +5 V for device type 02.
XTAL1, XTAL2	2,3	External clock source. See figures 4, 5, and 6.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Handling. MOS microprocessors should be handled with certain precautions to avoid damage due to the accumulation of static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Device type	Generic-industry type
01	8048
02	8035L

6.8 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

Custodian:
Air Force - 17
Navy - EC
Army - ER

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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